

# Control variates for a GARCH model

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*A Monte Carlo Study with Zero-Variance Estimators*

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1. Introduction
2. Model and Bayesian Setup
3. Sampling the Posterior
4. Variance Reduction: Zero-Variance Control Variates
5. Bonus: Validity of OLS on a Correlated Chain
6. Conclusion

PART 1

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# Introduction

# Motivation

## The core problem

Bayesian inference on GARCH models requires evaluating intractable posterior expectations, so we rely on **Markov Chain Monte Carlo (MCMC)**. But MCMC estimators can have **high variance**, requiring very long chains to be reliable.

- **Goal:** estimate the posterior mean of  $\theta = (\omega_1, \omega_2, \omega_3)$  in a Normal-GARCH(1,1) model efficiently.
- **Key idea:** replace  $f$  by a corrected estimator  $\tilde{f} = f + \mathbf{a}^T \mathbf{z}$  with the same expectation under  $\pi$  but much smaller variance, using **zero-variance control variates**.
- **Extension:** as the number of control variates grows, we use **LASSO** for dimensionality, and **subsampling** / **block averaging** to validate the regression step.

Based on Mira, Solgi & Imparato, *Stat. Comput.* 23 (2013), 653–662, applied to the DEM/GBP exchange rate data of Ardia (2008).

PART 2

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## **Model and Bayesian Setup**

# The Model — Normal-GARCH(1,1)

## Definition

The return process  $\{r_t\}$  follows

$$r_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, h_t), \quad h_t = \omega_1 + \omega_2 r_{t-1}^2 + \omega_3 h_{t-1}$$

with  $\omega_1 > 0$ ,  $\omega_2, \omega_3 \geq 0$ ,  $\omega_2 + \omega_3 < 1$  (covariance stationarity).

- $\omega_1$  — long-run variance floor (background volatility)
- $\omega_2$  — ARCH term: reaction to yesterday's shock
- $\omega_3$  — GARCH term: variance persistence (memory)
- Initialised at  $h_0 = \omega_1 / (1 - \omega_2 - \omega_3)$

# Bayesian Setup

## Prior

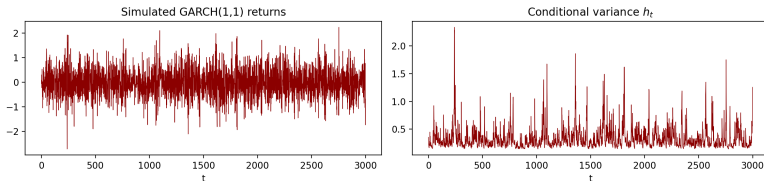
Independent truncated Gaussians:  $\omega_j \sim \mathcal{N}(0, 1) \mathbf{1}_{\omega_j \geq 0}$ ,  $j = 1, 2, 3$ . Weakly informative — posterior is driven by the data.

## Posterior

By Bayes' theorem:

$$\pi(\theta \mid r_{1:T}) \propto \underbrace{\exp\left(-\frac{1}{2} \sum_{t=1}^T \left[\log h_t + \frac{r_t^2}{h_t}\right]\right)}_{\text{likelihood}} \times \underbrace{\pi(\theta)}_{\text{prior}}$$

# The Simulated Data



$T = 3000$  returns simulated with  $\theta = (0.055, 0.245, 0.600)$ .

- Returns oscillate around zero (zero-mean by construction) with occasional spikes exceeding  $\pm 2.5\%$ .
- Visible **volatility clustering** — alternating calm and turbulent stretches, the GARCH signature.
- Conditional variance  $h_t$  never reaches zero (floor  $\sim 0.2$  from  $\omega_1$ ); spikes in  $h_t$  align with bursts in  $r_t$ .
- Unconditional std:  $\sqrt{0.055/(1 - 0.245 - 0.600)} \approx 0.6\%$  per day.

PART 3

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# **Sampling the Posterior**

# The Sampler — Random Walk Metropolis

## Why RWM?

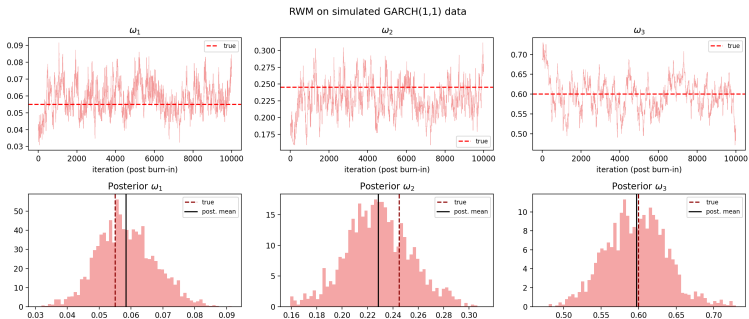
The posterior  $\pi(\theta \mid r) \propto L(\theta \mid r) \pi(\theta)$  has **no closed form** since the likelihood involves the recursive variance  $h_t$ . RWM only needs to *evaluate*  $\pi$  at two points per iteration.

- **Proposal:**  $\theta' = \theta + \varepsilon$ ,  $\varepsilon \sim \mathcal{N}(\mathbf{0}, \tau^2 I_3)$
- **Acceptance:**  $\min(1, \pi(\theta' \mid r) / \pi(\theta \mid r))$
- Computed in log-scale to avoid underflow
- Positivity enforced via  $\log \pi = -\infty$  outside the support

## Calibrating $\tau$

Aim for acceptance rate  $\approx$  **25%**: too high means the chain barely moves whereas too low means proposals are rejected and the chain freezes.

# RWM on Simulated Data

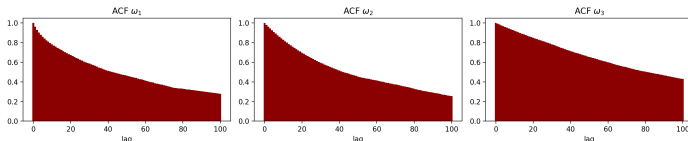


- $T = 3000$  simulated returns,  
 $\theta = (0.055, 0.245, 0.600)$
- $\tau = 0.01$ , acceptance = **27.1%** ✓
- $N = 12,000$ , burn-in = 2,000
- **Identification:**  $\omega_1$  &  $\omega_2$  partially compensate;  $\omega_3$  tightest.

|           | $\omega_1$ | $\omega_2$ | $\omega_3$ |
|-----------|------------|------------|------------|
| True      | 0.055      | 0.245      | 0.600      |
| Posterior | 0.0585     | 0.2285     | 0.5972     |

*Posterior recovery.*

# The Dependence Problem



*Autocorrelation decays slowly: ACF still  $\approx 0.4$  at lag 100.*

## Effective Sample Size

Out of 10,000 post-burn-in draws:

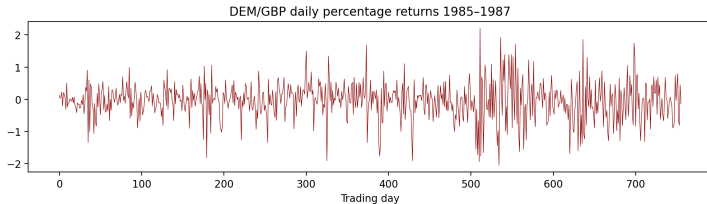
- ESS  $\omega_1 = 70$
- ESS  $\omega_2 = 76$
- ESS  $\omega_3 = 53$

$\sim 1$  independent sample / 140 iterations.

- Root cause: isotropic proposal does not adapt to the posterior ridge.
- MCMC variance  $\gg$  iid case  $\Rightarrow$  control variates are needed.

## Real Data — DEM/GBP 1985–1987

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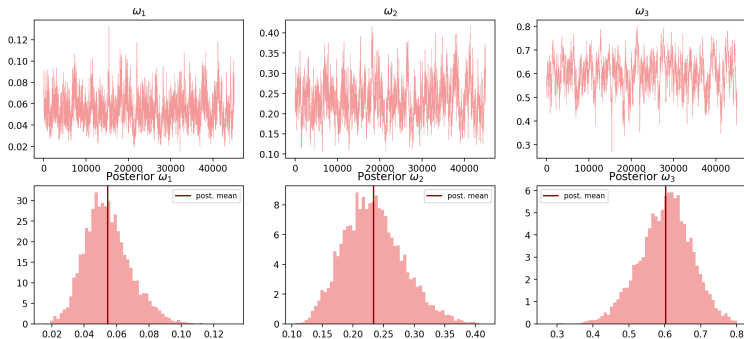


*756 daily returns of the DEM/GBP exchange rate (Ardia, 2008).*

- Same dataset used by Mira et al. (2013) for direct comparison.
- Visible volatility clustering: calm stretches alternating with turbulent ones — the empirical motivation for the GARCH model.
- Mean return  $\approx 0\%$ , std  $\approx 0.57\%$  per day.

# RWM on Real Data

RWM on DEM/GBP from 1985–1987



Trace plots and posterior histograms on 756 daily DEM/GBP returns.

- $\tau = 0.02$ , acceptance = **23.1%** ✓
- $N = 50,000$ , burn-in = 5,000
- $\hat{\omega}_2 + \hat{\omega}_3 \approx 0.84$ : stationary but persistent volatility

|       | $\omega_1$ | $\omega_2$ | $\omega_3$ |
|-------|------------|------------|------------|
| Ours  | 0.0545     | 0.2335     | 0.6027     |
| Paper | 0.055      | 0.245      | 0.600      |

**Implementation validated.**

PART 4

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# **Variance Reduction: Zero-Variance Control Variates**

# The Zero-Variance Principle (Mira et al. 2013)

## Idea

Replace  $f(\theta)$  with a corrected estimator  $\tilde{f}(\theta) = f(\theta) + \mathbf{a}^T \mathbf{z}(\theta)$ , where  $\mathbf{z}$  is **mean-zero** under  $\pi$ . Then  $\mathbb{E}_\pi[\tilde{f}] = \mathbb{E}_\pi[f]$ : same target but lower variance if  $\mathbf{a}$  is chosen well.

**Natural choice of  $\mathbf{z}$**  — the score:  $\mathbf{z}(\theta) = -\frac{1}{2} \nabla \log \pi(\theta \mid r)$ ,  $\mathbb{E}_\pi[\mathbf{z}] = \mathbf{0}$ .

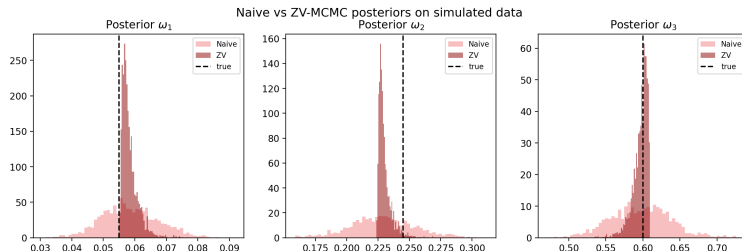
**Optimal  $\mathbf{a}$**  — found by **OLS regression** of  $f$  on  $\mathbf{z}$  along the chain:

$$f(\theta^i) = \alpha + \mathbf{a}^T \mathbf{z}(\theta^i) + \varepsilon_i$$

Intercept  $\hat{\alpha}$  is the ZV estimate of  $\mathbb{E}_\pi[f]$ ; residual variance gives  $\text{Var}(\tilde{f})$ .

For GARCH:  $\mathbf{z}$  computed by differentiating the variance recursion  $\dot{h}_t^{(j)} = \partial h_t / \partial \omega_j$  via the chain rule.

# Zero-Variance Degree-1 — Simulated Data

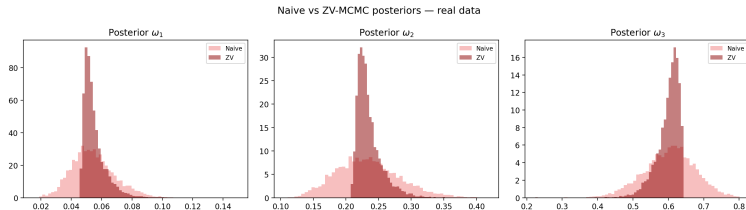


*Posteriors visibly tighten: the ZV correction collapses the bulk of the distribution.*

|            | MC std error (naive $\rightarrow$ ZV) | Reduction |
|------------|---------------------------------------|-----------|
| $\omega_1$ | 0.009 $\rightarrow$ 0.003             | 12.0×     |
| $\omega_2$ | 0.025 $\rightarrow$ 0.005             | 22.2×     |
| $\omega_3$ | 0.040 $\rightarrow$ 0.010             | 15.4×     |

*Std errors in units of  $10^{-2}$ . All within the 95% CI of Mira et al. (2013).*

# Zero-Variance Degree-1 — Real Data (DEM/GBP)



Same pattern, smaller magnitude: the linear  $\mathbf{a}^T \mathbf{z}$  explains less of the irregular real-data posterior.

|            | MC std error (naive $\rightarrow$ ZV) | Reduction |
|------------|---------------------------------------|-----------|
| $\omega_1$ | 0.007 $\rightarrow$ 0.004             | 3.6×      |
| $\omega_2$ | 0.023 $\rightarrow$ 0.009             | 6.1×      |
| $\omega_3$ | 0.034 $\rightarrow$ 0.016             | 4.8×      |

**Real data underperforms**  $\Rightarrow$  motivates degree-2.

# Scaling Up — Degree-2 Polynomials

## The 9 control variates

With  $P(\theta) = \mathbf{a}^T \theta + \frac{1}{2} \theta^T B \theta$ :

$$\tilde{f}(\theta) = f(\theta) - \frac{1}{2} \text{tr}(B) + (\mathbf{a} + B\theta)^T \mathbf{z}(\theta)$$

- 3 linear:  $z_1, z_2, z_3$
- 6 quadratic:  $\omega_j z_k$  for  $j \leq k$

## Why not just OLS?

Cost  $\mathcal{O}(nm^2 + m^3)$ ,  $m = \frac{d(d+3)}{2}$ :

| Degree | $d = 3$ | $d = 10$ | $d = 50$ |
|--------|---------|----------|----------|
| 1      | 3       | 10       | 50       |
| 2      | 9       | 65       | 1325     |
| 3      | 19      | 285      | 23,425   |

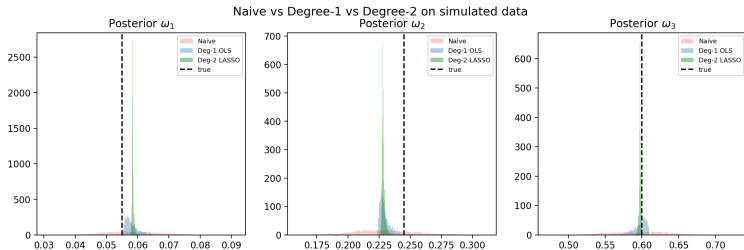
Prohibitive at moderate  $d$ .

## LASSO (Leluc et al. 2021)

$\ell_1$  penalty:  $\min_{\mathbf{a}} \|f - H\mathbf{a}\|^2 + \lambda \|\mathbf{a}\|_1$

- Selects relevant variates
- $\lambda$  tuned by 5-fold CV
- Avoids overfitting on autocorrelated draws
- Robust at large  $m$

# Degree-2 LASSO — Simulated Data

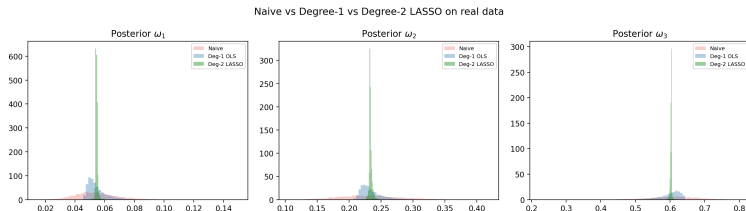


*Degree-2 LASSO (green) posteriors collapse onto the true values (dashed) — invisibly thin in the histogram.*

| Method      | $\omega_1$  | $\omega_2$  | $\omega_3$   |
|-------------|-------------|-------------|--------------|
| Naive       | 1×          | 1×          | 1×           |
| Deg-1 OLS   | 12×         | 22×         | 15×          |
| Deg-2 LASSO | <b>738×</b> | <b>422×</b> | <b>1028×</b> |

*LASSO selects 9/9 ( $\omega_1, \omega_3$ ) and 8/9 ( $\omega_2$ ) variates — no sparsity at  $d = 3$ .*

# Degree-2 LASSO — Real Data (DEM/GBP)



*Same dramatic collapse on real financial data.*

| Method      | $\omega_1$  | $\omega_2$  | $\omega_3$  |
|-------------|-------------|-------------|-------------|
| Naive       | 1×          | 1×          | 1×          |
| Deg-1 OLS   | 3.6×        | 6.1×        | 4.8×        |
| Deg-2 LASSO | <b>298×</b> | <b>340×</b> | <b>760×</b> |

**Three orders of magnitude** reduction: it confirms the paper's central finding.

PART 5

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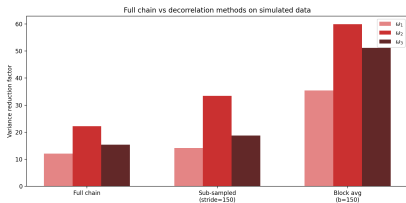
## **Bonus: Validity of OLS on a Correlated Chain**

# Bonus — Is OLS Valid on a Correlated Chain?

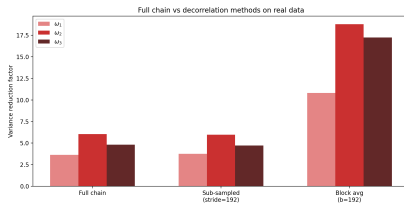
## The issue

OLS assumes **i.i.d.** observations. With autocorrelated MCMC draws:  $\hat{\Sigma}_{zz}$  uses fewer effective samples than  $N$ , coefficients  $\hat{\mathbf{a}}$  are less precise, and LASSO cross-validation underestimates uncertainty (adjacent folds nearly identical).

- **Sub-sampling**: keep every  $n/\text{ESS}$ -th draw
- **Block averaging**: replace each block by its mean



*Simulated data — both methods help, block avg triples the reduction.*



*Real data — sub-sampling barely helps, only block averaging delivers.*

**Conclusion:** block averaging is the robust decorrelation strategy. Moreover, the Q2/Q3 results were *conservative*, not optimistic.

PART 6

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# Conclusion

## Conclusion

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- Implemented Bayesian Normal-GARCH(1,1) estimation via RWM with posterior means that do match Mira et al. (2013) within 1%.
- **Degree-1 ZV control variates:** 12–22 $\times$  variance reduction (simulated), 4–6 $\times$  (real).
- **Degree-2 + LASSO: three orders of magnitude** reduction ( $\sim 10^3\times$ ), confirming the paper's finding and extending it via Leluc et al. (2021).
- Addressed OLS validity on correlated MCMC: block averaging shows our reductions were **conservative**.

### Possible extensions

Higher-degree polynomials, adaptive MCMC for better mixing, multivariate or asymmetric GARCH (E-GARCH, GJR-GARCH).

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*Thank you.*

# References I

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- [1] Mira, A., Solgi, R., & Imparato, D. (2013). *Zero variance Markov chain Monte Carlo for Bayesian estimators*. *Statistics and Computing*, 23(5), 653–662.
- [2] Leluc, R., Portier, F., & Segers, J. (2021). *Control variate selection for Monte Carlo integration*. *Statistics and Computing*, 31(4), 1–26.
- [3] Bollerslev, T. (1986). *Generalized autoregressive conditional heteroskedasticity*. *Journal of Econometrics*, 31(3), 307–327.
- [4] Ardia, D. (2008). *Financial Risk Management with Bayesian Estimation of GARCH Models*. Springer.